DP Biology – A1.1

Water

- Which features of water make life possible?
- What does living in water make easier and harder for organisms?

SL statements A1.1.1–6. Textbook pp. 2–10.
Image: Earth from Apollo 17. NASA, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	A1.1.1–4	Explain how polar covalent bonds lead to hydrogen bonds, and how cohesion and adhesion matter to organisms
2	A1.1.5–6	Explain why water is a good solvent, and compare the physical properties of water and air for animals

One property explains the rest



Electrons in a water molecule are shared **unequally**, so each molecule is slightly charged at its ends. Neighbouring molecules attract each other by **hydrogen bonds**.

Those attractions explain why water sticks to itself and to surfaces, dissolves so much, and heats up slowly.

Big question

Why do scientists searching for life on other planets look for liquid water first?

Water is the medium for life

A1.1.1

The **first cells** arose in water, and most life processes still happen in it.

Water makes up most of the **cytoplasm**, **tissue fluid** and **blood plasma**, so its properties affect everything cells do.

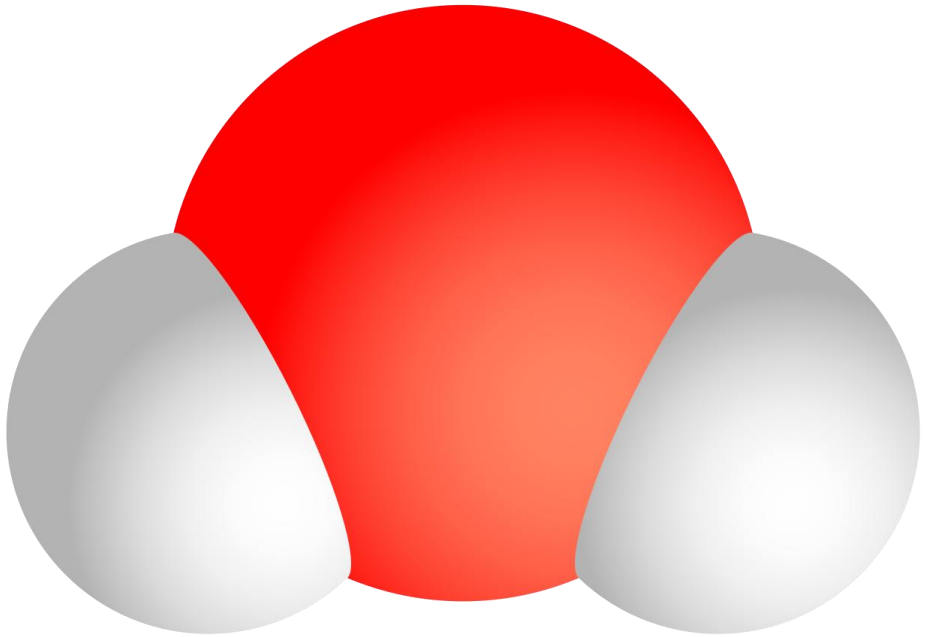
Organism or part	Water
Human body	about 60%
Cytoplasm	70–80%
Lettuce leaf	about 95%
Dry seed	about 10%

Think

A dry seed is alive but almost inactive. Why can’t it germinate until it takes up water?

Polar covalent bonds

A1.1.2



A water molecule. Sakurambo, public domain.

Oxygen shares a pair of electrons with each hydrogen: **covalent bonds**. Oxygen pulls the electrons closer, so it is slightly negative (δ−) and each hydrogen slightly positive (δ+).

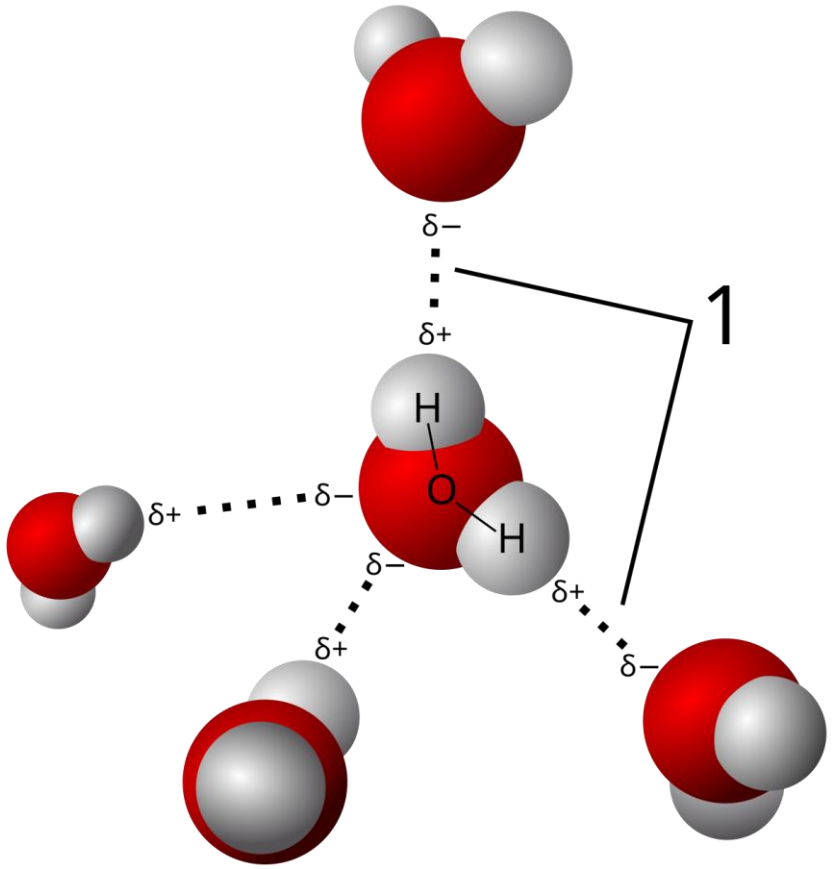
The molecule is **bent**, so the charges do not cancel: water is **polar**.

Notation

Show partial charges with the Greek letter delta: δ+ on each hydrogen, δ− on the oxygen.

Hydrogen bonds between water molecules

A1.1.2



Hydrogen bonds in water. Qwerter, public domain.

The δ+ hydrogen of one molecule is attracted to the δ− oxygen of a neighbour: a **hydrogen bond**.

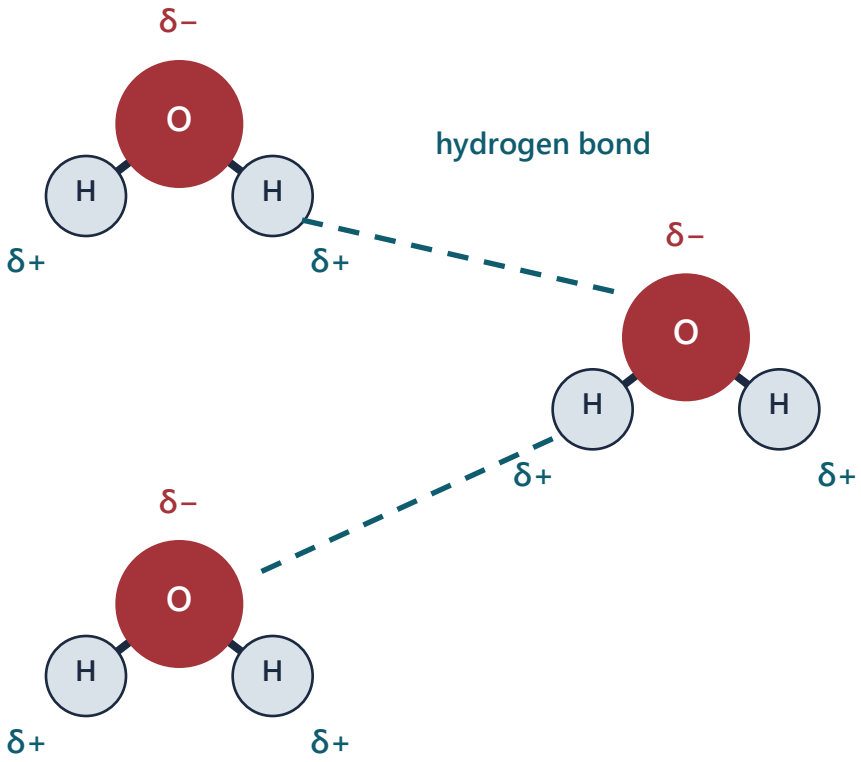
Each one is weak and keeps breaking and re-forming, but there are so many that together they give water its properties. Each molecule can form up to four.

Common mistake

A hydrogen bond is between molecules. The bonds inside a water molecule are covalent.

How to draw hydrogen bonds

A1.1.2



Bent molecules; δ− on each O and δ+ on each H.

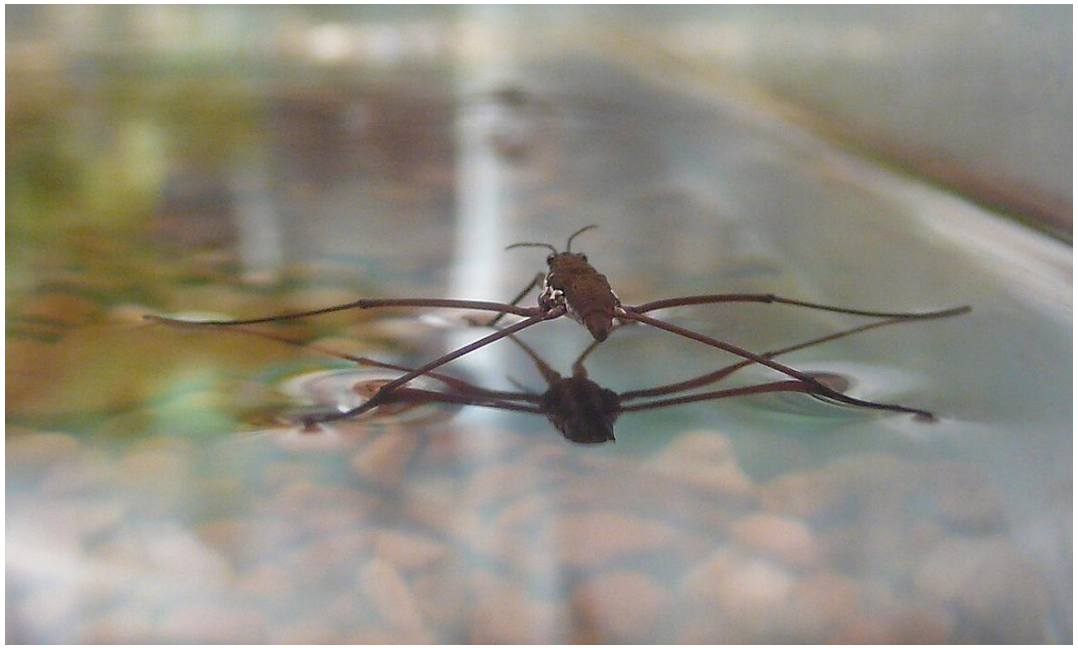
Hydrogen bonds as **dashed lines** from an H of one molecule to the O of another; covalent bonds solid.

Try it

Draw four water molecules with at least three hydrogen bonds; check a partner's against the rules.

Cohesion: water sticks to itself

A1.1.3



A water strider on the surface. TimVickers, public domain.

Cohesion is the attraction of water molecules to each other by hydrogen bonds.

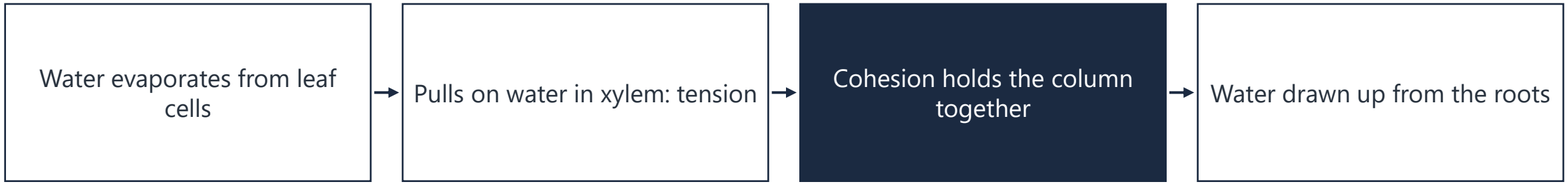
At the surface this gives **surface tension**: small animals such as water striders can stand on it.

Key idea

Cohesion resists pulling, so a thin column of water can be pulled up a plant without breaking.

Cohesion lets plants pull water up

A1.1.3



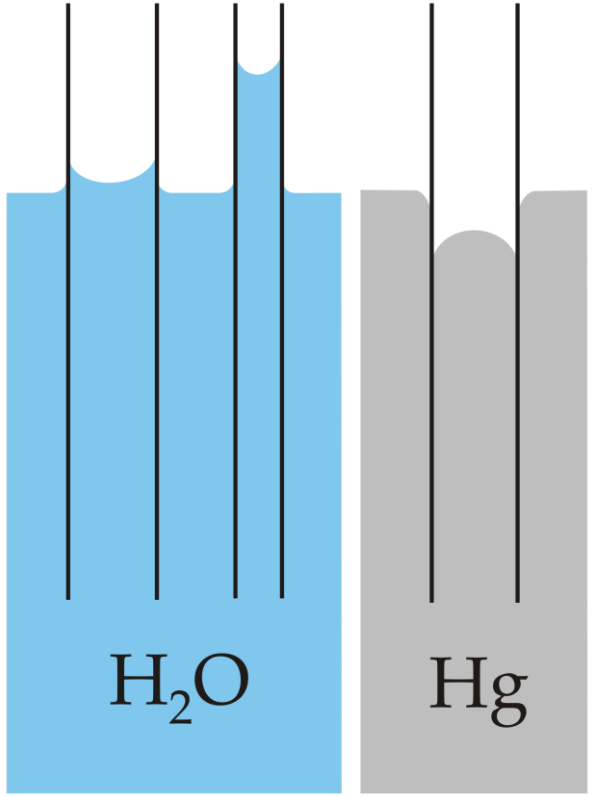
Evaporation from leaves pulls the water in the xylem. Because of **cohesion**, the pull passes down the whole column without breaking it, even up trees over 100 m tall.

Link

Transport in plants (B3.2): adhesion to xylem walls also helps.

Adhesion: water sticks to other surfaces

A1.1.4



Capillary action. MesserWoland, CC BY-SA 3.0.

Adhesion is water's attraction to polar or charged surfaces such as cellulose, glass or clay.

Adhesion pulls water up the sides of a narrow tube; cohesion pulls the rest after it: **capillary action**.

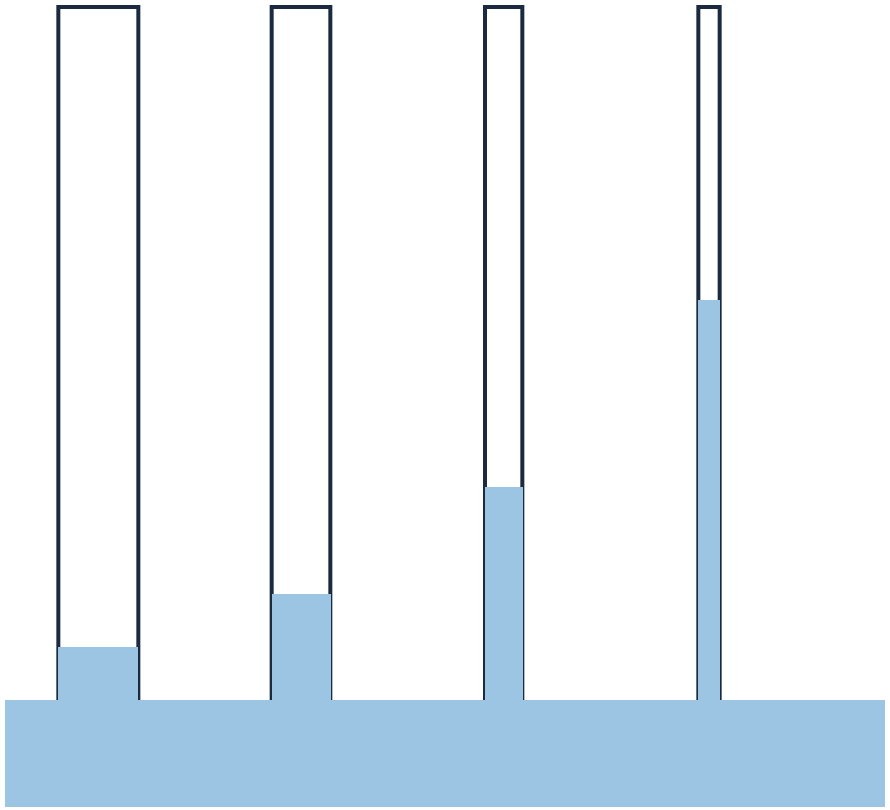
Where	What capillary action does
Soil	Draws water up between particles towards roots
Cell walls	Keeps the walls of leaf cells wet

Common mistake

Cohesion: water to water. Adhesion: water to something else.

Working with data: capillary rise

A1.1.4



drawn tubes, narrowest on the right

Tube width / mm	Rise / mm
2.0	15
1.0	30
0.5	60
0.25	120

1. Describe the trend.
2. Predict the rise in a 0.1 mm tube.
3. Explain using adhesion.

Check-in: class 1

A1.1.1–4

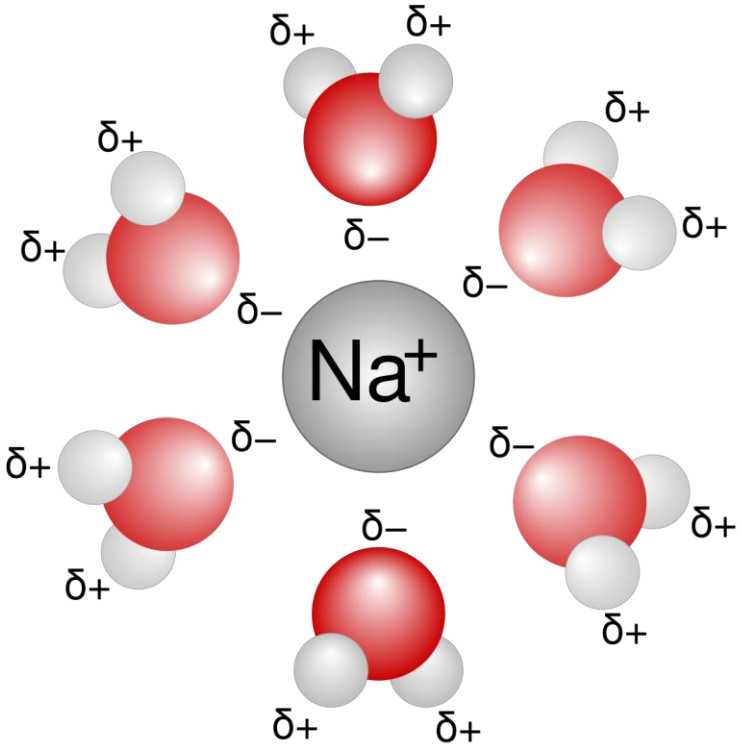
1. Explain why a water molecule is polar.
2. Distinguish between a covalent bond and a hydrogen bond in water.
3. Draw two water molecules joined by a hydrogen bond.
4. Explain how cohesion helps water move up a tree.
5. Give one example of adhesion that matters to plants.

How to use this

Answer without your notes, then check against slides 3–11.

Water as a solvent

A1.1.5



Water around a sodium ion. Steven P. Berg, public domain.

Polar water molecules cluster around **ions** and **polar molecules**, separating them so they dissolve. Around Na^+ the δ− oxygens face in; around a chloride ion the δ+ hydrogens face in.

Substances that dissolve easily are **hydrophilic**: salts, sugars, amino acids.

Why it matters

Water carries dissolved substances around organisms and brings them together to react.

Water is the medium for metabolism

A1.1.5

Most cell reactions happen in **aqueous solution**, where molecules move freely and meet enzymes.

Water is also a reactant or product: used in **hydrolysis**, released in **condensation**.

Reaction	Water
Digesting starch	Used
Making a protein	Released
Photosynthesis	Used
Aerobic respiration	Released

Link

B1.1 and B1.2: condensation and hydrolysis.

How blood carries substances

A1.1.5

Substance	In water	How blood carries it
Sodium chloride	Soluble	Dissolved as ions in plasma
Glucose	Soluble	Dissolved: -OH groups bond with water
Amino acids	Soluble	Dissolved in plasma
Oxygen	Poorly soluble	Bound to haemoglobin in red cells
Fats, cholesterol	Insoluble	Inside lipoprotein particles

Key idea

Insoluble substances must be packaged or bound to proteins.

When being insoluble is useful

A1.1.5



Water droplets on a waxy leaf. Siddharth Patil, CC0.

Hydrophobic molecules are non-polar and do not dissolve. Many work **because** of this.

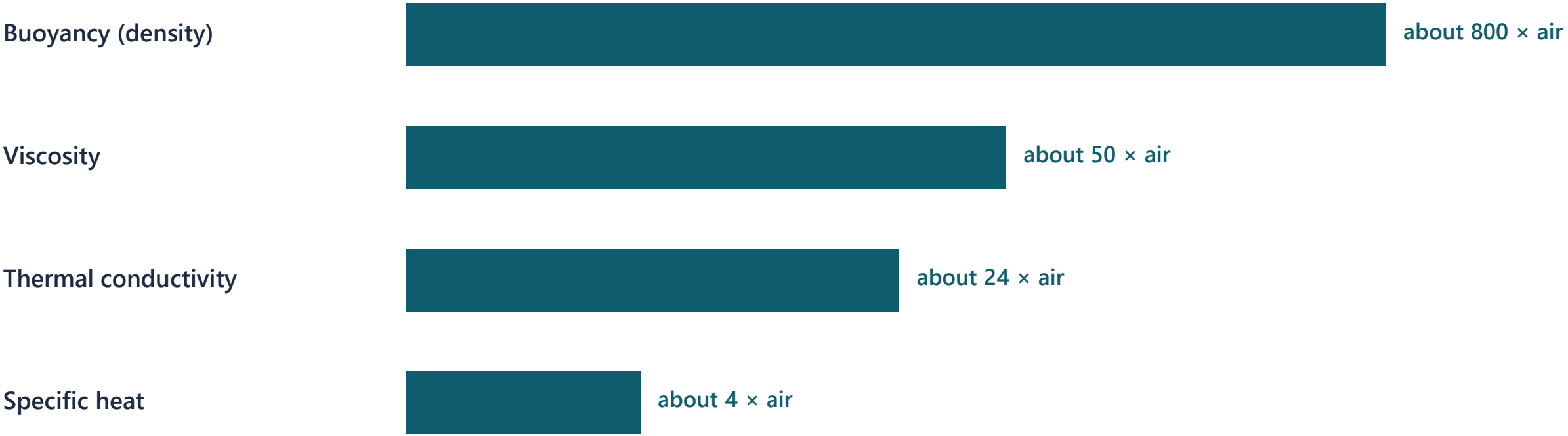
Membranes' hydrophobic cores block ions; fats store energy without affecting water balance; a waxy cuticle cuts water loss.

Common mistake

Water does not repel hydrophobic molecules: water molecules attract each other more, so the hydrophobic ones are pushed together.

Water compared with air

A1.1.6



Drawn on a log scale. Approximate values at 20 °C.

Key idea

High specific heat comes from hydrogen bonds: energy breaks them before molecules speed up.

What these properties mean for animals

A1.1.6

Property	Challenge or opportunity	Adaptation
Buoyancy	Supports large bodies; staying level costs energy	Swim bladder; fat or oil
Viscosity	Drag when moving	Streamlined body
Thermal conductivity	Warm-blooded animals lose heat fast	Blubber; waterproof feathers
Specific heat	Stable water temperature	Narrow tolerance in many species

Think

Why can the blue whale only live in water?

Water and air: the seal and the loon

A1.1.6



Ringed seal pup. Shawn Dahle, NOAA, public domain.



Black-throated loon. Robert Bergman, public domain.

	Ringed seal	Black-throated loon
Buoyancy	Blubber helps it float	Dense bones; squeezes air out to dive
Viscosity	Streamlined; flippers	Torpedo shape; webbed feet set back
Heat loss	Thick blubber	Dense, oiled feathers trap air
In air	Slow on ice	Must fly: can’t be too heavy

Key idea

Living in both water and air needs compromises.

Working with data: losing heat

A1.1.6

Flasks of water at 40 °C as model animals, after 20 minutes. Some wrapped in 1 cm of lard (model blubber). Values invented for practice; realistic pattern.

Flask	Start / °C	After 20 min
In air at 10 °C	40	33
In water at 10 °C	40	21
In water, with lard	40	31

1. Calculate each temperature fall.
2. Explain why water cooled the flask faster.
3. Explain the effect of the lard.
4. Suggest one improvement.

Check-in: class 2

A1.1.5–6

1. Explain why salt dissolves in water.
2. Explain how blood carries oxygen and fats.
3. Give two examples of molecules that work because they are hydrophobic.
4. Explain why water has a high specific heat.
5. Describe two adaptations of the ringed seal for life in water.

How to use this

Answer without your notes, then check against slides 13–20.

Common mistakes to avoid

Not this	But this
Hydrogen bonds hold a water molecule's atoms together.	Covalent bonds do; hydrogen bonds are between molecules.
Water is polar because it contains oxygen.	It is polar because sharing is unequal and the molecule is bent.
Cohesion and adhesion are the same.	Cohesion: water to water. Adhesion: water to another surface.
Water heats slowly because it is dense.	Energy is used to break hydrogen bonds.

Summary: water

A1.1.1–6

	Key idea
A1.1.1	Life began in water; cells are mostly water
A1.1.2	Polar covalent bonds; hydrogen bonds between molecules
A1.1.3	Cohesion: surface tension, xylem
A1.1.4	Adhesion: capillary action in soil and cell walls
A1.1.5	Solvent for metabolism and transport; hydrophobic uses
A1.1.6	Buoyancy, viscosity, conductivity, specific heat; seal and loon

Check that you can...

1. **Explain** why water molecules are polar, and **draw** hydrogen bonds
2. **Explain** cohesion and adhesion, with examples
3. **Explain** why water is a good solvent
4. **Outline** why some molecules work because they are hydrophobic
5. **Compare** the physical properties of water and air
6. **Discuss** the adaptations of the ringed seal and loon

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why water molecules are polar.
2. Distinguish between cohesion and adhesion, giving one example of each in organisms.
3. Explain why a seal needs a thick layer of blubber in cold water.

Image credits

The Earth seen from Apollo 17.jpg – NASA/Apollo 17 crew; taken by either Harrison Schmitt or Ron – Public domain

Water molecule.svg – Sakurambo – Public domain

3D model hydrogen bonds in water.svg – User Qwerter at Czech wikipedia: Qwerter. Transferred from c – Public domain

Water strider.jpg – TimVickers – Public domain

Capillarity.svg – MesserWoland – CC BY-SA 3.0

Na+H2O.svg – Original: Dr. Steven P. Berg Vector: Duncan Keall, Glrx, JoK – Public domain

Water droplets on leaf.jpg – Siddharth Patil – CC0

Pusa hispida pup.jpg – Shawn Dahle, NOAA, Polar Ecosystems Program research cruise – Public domain

Gavia arctica1.jpg – Robert Bergman – Public domain

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses